EXPERIMENT NO. 6
RESONANCE TUBE

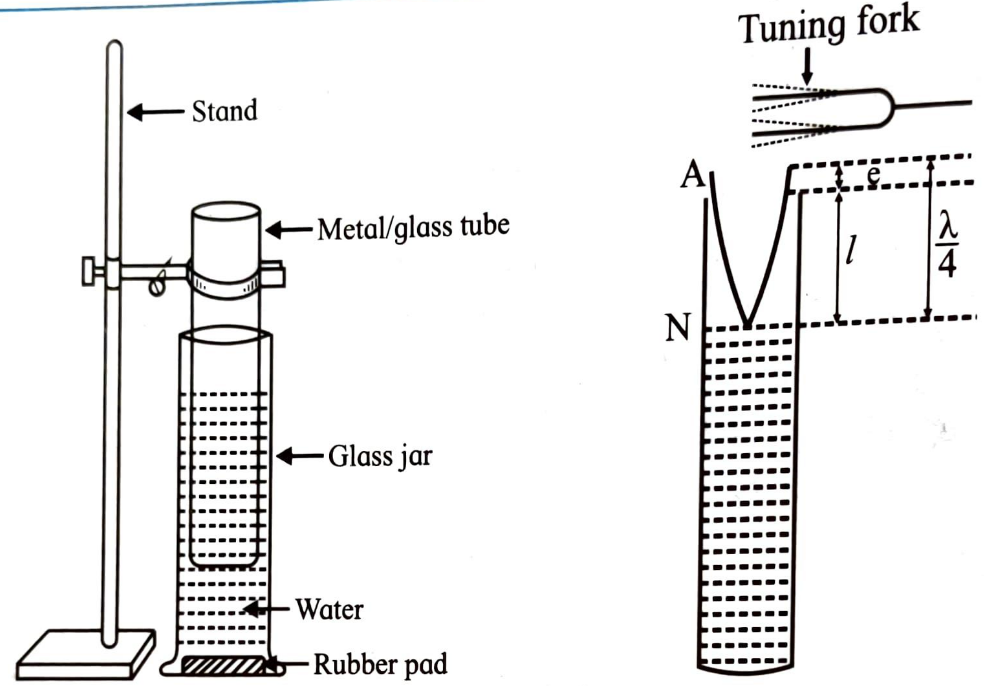
Aim : (i) To determine the speed of sound in air at room temperature

(ii) To find out the unknown frequency of a tuning fork using Resonance Tube

Apparatus :

A long plastic jar filled with water, resonance tube, set of tuning forks, retort stand, meter scale, vernier calliper, rubber pad, thermometer

Diagram :



Formula :

(i) End correction $e = 0.3d$ where d is the inner diameter of resonance tube

(ii) $V = 4\ (nL)_{mean}$ n= frequency of tuning fork

$= 4n\ (\ell + e)$ L = correct resonating length of air column

$= 4n\ (\ell + 0.3d)$ ℓ = observed resonating length of air column

(iii) $V = 4 \times \dfrac{1}{\text{slope}}$ (if graph is $\dfrac{1}{L}$ against n)

(iv) $n_x = \dfrac{(nL)\,\text{mean}}{L_x}$ (where L_x is the correct resonating length for fork of unknown frequency)

Procedure :

1. Measure the inner diameter of the resonance tube with the help of the upper jaws of vernier calliper. Take three readings and hence calculate end correction of resonance tube.
2. Note down the room temperature using the given thermometer.
3. Hold the resonance tube vertically in the plastic jar filled with water with the help of retort stand (as shown in the diagram).

4. Immerse the resonance tube in the plastic jar filled with water as deep as possible so that the length of the air column is minimum.
5. Arrange the tuning forks in descending order of frequency.
6. Strike the tuning fork of highest frequency gently on the rubber pad and hold it near the mouth of the resonance tube, so that the prong of the tuning fork vibrate in vertical plane.
7. Raise the resonance tube slowly along with the fork till a loud sound is heard indicating that the air column is vibrating in resonance with the fork. Clamp the tube.
8. Measure the length of air column from the surface of the water level to the open end of the resonance tube using meter scale
9. Repeat the procedure and measure the length (ℓ) three times for the same tuning fork and calculate mean length (ℓ)
10. Set the next tuning fork in to vibrations, hold it near the open end of the resonance tube, raise it slowly from the previous position till resonance is obtained and measure the corresponding length (ℓ) three times and calculate the mean length
11. Repeat the complete procedure for the rest of the tuning forks
12. Similarly find the mean resonating length (ℓ_x) for the unknown frequency tuning fork (n_x)
13. Add end correction (e) to mean resonating length (ℓ) and calculate corrected length (L) for all tuning forks. Calculate L_x for the tuning fork of unknown frequency
14. Calculate nL for all tuning forks except the tuning fork of unknown frequency
15. Calculate the speed of sound at room temperature and also unknown frequency
16. Plot a graph of n (y axis) against $\frac{1}{L}$ (x axsis) and calcualte the speed of sound by graph.

Observations :

1. Least count of vernier calliper

(i) Smallest division on the main scale of vernier calliper, x = ... cm

(ii) Total number of divisions on the vernier scale of the vernier calliper, y = ... div.

(iii) Least count (L.C) of vernier calliper, x/y = ... cm

(iv) Zero error (e') = ... cm

Observation table :

1) For inner diameter of resonance tube

Obs. No.	MSR / (a)cm	VSD (b)cm	VSR (c= b x L.C) cm	Diameter / (total reading) (a+c = d') cm	Corrected reading (d = d'− e') cm
1	2.4	8	0.08	2.48	
2	2.4	8	0.08	2.48	2.51
3	2.5	9	0.09	2.59	

Mean d = ___ cm

End correction e = 0.3 d = ____ cms

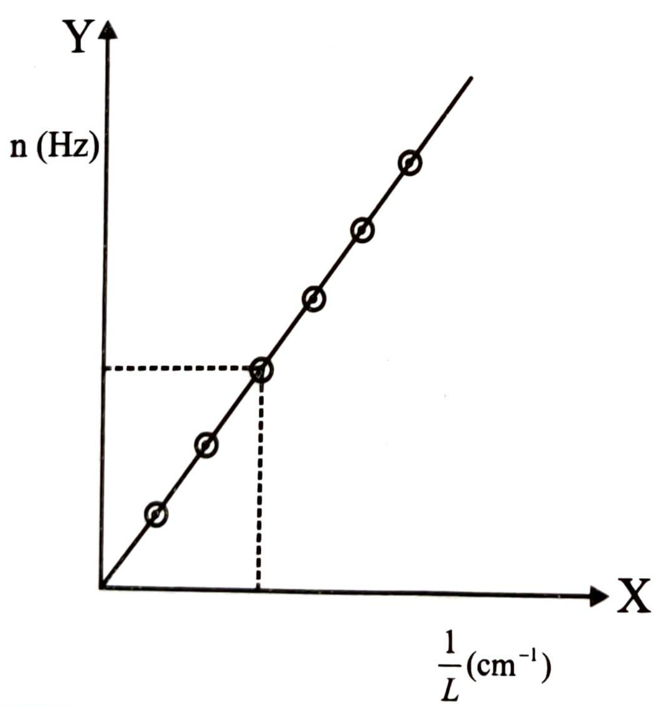
2) For resonating length :

Obs. No.	Frequency n (Hz)	Length of air column				Corrected length $L = \ell + e$ cm	$\frac{1}{L}$ (cm-1)	nL Hz - cm
		ℓ_1 cm	ℓ_2 cm	ℓ_3 cm	Mean ℓ cm			
1	320 Hz	26.5	25.5	25.4	25.8	25.8	$\frac{1}{25.8}$	8256
2	384 Hz	20.7	20.4	20.3	20.46	20.46	$\frac{1}{20.46}$	7856.64
3	426 Hz	16.4	16.0	15.8	16.06	16.06	$\frac{1}{16.06}$	6841.56
4	480 Hz	15.4	15.4	15	15.26	15.26	$\frac{1}{15.26}$	7324.8
5	Unknown $(n_x) =$					$L_x =$		$\frac{1}{L_x} =$

Mean (nL) = 7569.75 Hz – cm or cm/s

Graph :

Plot a graph of n (y axis) against $\frac{1}{L}$ (x axis) and calculate the speed of sound by graph.



Scale X = ---------------

Y = ---------------

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

V = 4 × slope = ____ cm/s = ____ m/s

Also determine the unknown frequency (n_x) from the graph

1] $8 \times 0.01 = 0.08$
2] $9 \times 0.01 = 0.09$
3] $2.4 + 0.08 = 2.48$
4] $2.4 + 0.08 = 2.48$
5] $2.5 + 0.09 = 2.59$
6] $\frac{2.48 + 2.48 + 2.59}{3} = 2.51$

7] $e = 0.3d = 0.3 \times 2.51 = 7.53$
8] $\frac{26.5 + 25.5 + 25.4}{3} = 25.8$

9] $\frac{20.7 + 20.4 + 20.3}{3} = 20.46$

10] $\frac{16.4 + 16.0 + 15.8}{3} = 16.06$

11] $\frac{15.4 + 15.2 + 15}{3} = 15.26$

12] $25.8 + 7.53 = 33.33$
13] $20.5 + 7.53 = 28.03$
14] $16.2 + 7.53 = 23.73$
15] $15.2 + 7.53 = 22.73$

16] $320 \times 25.8 = 8256$
17] $384 \times 20.46 = 7856.64$
18] $426 \times 16.06 = 6841.56$
19] $480 \times 15.26 = 7324.8$.

Log n = a.

1) 320 = 2.5051
2) 384 = 2.5843
3) 426 = 2.6294
4) 480 = 2.6812

a + b = c.

1) 2.5051 + 1.4116 = 3.9167
2) 2.5843 + 1.3109 = 3.8952
3) 2.6294 + 1.2057 = 3.8351
4) 2.6812 + 1.1835 = 3.8647

Log d = b.

1) 25.8 = 1.4116
2) 20.46 = 1.3109
3) 16.06 = 1.2057
4) 15.26 = 1.1835

Antilog c = And.

1) $3.9167 = 8.252 \times 10^3$
2) $3.8952 = 7.856 \times 10^3$
3) $3.8351 = 6.841 \times 10^3$
4) $3.8647 = 7.324 \times 10^3$.

Calculations :

Obs. No	Log n (a)	Log L (b)	a + b = c	Antilog (c) = 4nL
1	2.5051	1.4116	3.9167	8.252×10^3
2	2.5843	1.3109	3.8952	7.856×10^3
3	2.6294	1.2057	3.8351	6.841×10^3
4	2.6812	1.1835	3.8647	7.324×10^3

Mean (nL) = 7569.75 Hz cm or cm / s

(1) Speed of Sound

$V = 4(nL)_{mean}$

$= 4 \times$ 7569.75

= _____ Hz cm or cm / s = ____ m/s

(2) Unknown frequency

$$n' = \frac{\text{mean (nL)}}{\text{Corrected length corresponding to the unknown frequency } (L_x)}$$

= ____ Hz

Result :

1) Speed of sound in air at room temperature
   a) By calculation : V = ____ m/s
   b) By graph : V = ____ m/s

2) Unknown frequency of fork
   a) By calculation : n' = ____ Hz
   b) By graph : n' = ____ Hz

Precautions:

1. Strike the tuning fork gently on rubber pad
2. Hold the vibrating tuning fork just above the mouth of the resonance tube
3. Do not bring ear close to the tube
4. While adjusting the resonating length, start with the minimum length of the air column and the tuning fork of highest frequency
5. Tuning fork should not touch the resonance tube
6. The prongs of the tuning fork must vibrate in vertical plane

Questions

1. What is meant by the term resonance ?

Resonance is the phenomenon in which a body vibrates with maximum amplitude when the frequency of an external periodic force becomes equal to its natural frequency.

2. What are forced oscillations ?

The oscillations of a body under the influence of an external periodic force are called forced oscillations.

3. What are the sources of errors in this experiment ?

Sources of errors include inaccurate measurement of length, tension and frequency, friction at the pulley, improper tuning of the tuning fork, and errors in observation.

4. What are nodes and antinodes ?

A node is a point on a stationary wave where the amplitude is zero, while an antinode is a point where the amplitude is maximum.

5. What is end correction ? How do you eliminate it ?

End correction is the difference between the effective length and the actual length of the air column in a resonance tube. It can be eliminated by taking two successive resonance lengths and using their difference.

6. State the factors on which velocity of sound depends ?

The velocity of sound depends on the nature of the medium, its temperature, humidity and pressure.

7. How does the speed of sound in the given medium vary with the temperature ?

In a given medium, the speed of sound increases with increase in temperature.

Remark and sign of teacher : ....

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